

The Bias Amplification Paradox in Text-to-Image Generation

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Abstract

Bias amplification is a phenomenon in which models exacerbate biases or stereotypes present in the training data. In this paper, we study bias amplification in the text-to-image domain using Stable Diffusion by comparing gender ratios in training vs. generated images. We find that the model appears to amplify gender-occupation biases found in the training data (LAION) considerably. However, we discover that amplification can be largely attributed to discrepancies between training captions and model prompts. For example, an inherent difference is that captions from the training data often contain explicit gender information while our prompts do not, which leads to a distribution shift and consequently inflates bias measures. Once we account for distributional differences between texts used for training and generation when evaluating amplification, we observe that amplification decreases drastically. Our findings illustrate the challenges of comparing biases in models and their training data, as well as evaluation more broadly, and highlight how confounding factors can impact analyses.

1 Introduction

Breakthroughs in machine learning have been fueled in large part by training models on massive unlabeled datasets such as the Pile, C4, and LAION (Gao et al., 2020; Raffel et al., 2020; Schuhmann et al., 2022). However, several studies have shown that these datasets exhibit biases and undesirable stereotypes (Birhane et al., 2021; Dodge et al., 2021; Garcia et al., 2023), which in turn impact model behavior. Given that models are trained to represent the data distribution, it is not surprising that models perpetuate biases found in the training data (De-Arteaga et al., 2019; Sap et al., 2019; Adam et al., 2022, among others).

To introduce bias amplification, let us take a model that generates images of engineers that are female 10% of the time using a gender-neutral



Figure 1: Comparing generated and training images for **engineer**, the model clearly seems to amplify bias by generating 10% female images, as compared to 25% female in training images. However, when looking at the subset of training examples *without gender indicators* in captions (10% female), similar to our prompts, the model does not amplify bias.

prompt. When examining the training data, we may assume that the model reflects associations in the data and expect to observe roughly 10% female as well.¹ However, it would be problematic for the model to instead exacerbate existing imbalances by generating engineer images that are only 10% female, while the training engineer images are 25% female, as shown in Figure 1. This phenomenon, known as *bias amplification* (Zhao et al., 2017), is concerning because it further reinforces stereotypes and widens disparities. While previous works suggest that models amplify biases (Zhao et al., 2017; Wang et al., 2018; Hall et al., 2022; Hirota et al., 2022; Friedrich et al., 2023), there remain unanswered questions about the paradoxical nature of bias amplification, given below:

¹Note that even such bias *preservation* may be undesirable.

The Bias Amplification Paradox

Given that models learn to fit the training data by maximizing its likelihood, why do models amplify biases found in the data as opposed to strictly representing them?

In this paper, we investigate how model biases compare with biases found in the training data. We focus on the text-to-image domain in English and analyze gender-occupation biases in Stable Diffusion (Rombach et al., 2022), as well as its publicly available training dataset LAION (Schuhmann et al., 2022), which consists of image-caption pairs (§2). To select training examples, we identify captions that mention occupations (e.g., engineer) and obtain their corresponding images. We follow previous work (Bianchi et al., 2023; Luccioni et al., 2023) and use prompts that contain a given occupation (e.g., “A photo of the face of an engineer”) to generate images. For each occupation, we then classify binary gender to measure bias in corresponding training and generated images, and compare the respective quantities to determine whether the model amplifies biases² from its training data (§3).

At first glance, it appears that the model amplifies bias considerably (on average, generation bias is 12.57% higher than training bias) using existing approaches (§4). When comparing training captions and prompts, however, we discover clear distributional differences that impact amplification measurements. For example, one inherent distinction is that captions often specify explicit gender indicators while prompts used to study gender-occupation biases do not.³ More generally, captions may contain additional context and details that are absent from the prompts we use.

Based on our observations, it is clear that directly using all training captions that mention a given occupation provides a naive characterization of bias amplification. Instead, we propose evaluating amplification on subsets of the training data that reduce distribution shifts between training and generation (§5). We introduce two approaches to account for distributional differences: (1) Excluding captions with explicit gender information and (2) Using nearest neighbors (NN) on text embeddings

to select training captions that closely resemble prompts. Both approaches restrict the search space of training texts to more closely match prompts, which results in considerably lower amplification measures. We then eliminate differences between training captions and prompts by utilizing the captions themselves to generate images (§6), and show that amplification is minimal. By modifying subsets of captions and prompts used to evaluate amplification, we perform a multi-pronged analysis of distribution shifts that impact evaluation.

To summarize, we study gender-occupation bias amplification in Stable Diffusion and highlight notable discrepancies between texts used for training and generation. We demonstrate that naively quantifying bias provides an incomplete and misleading depiction of model behavior. Our work emphasizes that comparisons of dataset and model biases should factor in distributional differences and evaluate comparable distributions. We hope that our work encourages future studies that analyze model behavior through the lens of the data.

2 Experimental Setup

Before discussing how we define and evaluate amplification in the following section, we first outline the dataset and models in our experiments, as well as how we infer gender from images.

2.1 Dataset and Models

To study bias amplification, we use Stable Diffusion (Rombach et al., 2022), a text-to-image model that generates images based on a textual description (prompt). Stable Diffusion is trained on pairs of captions and images taken from LAION-5B (Schuhmann et al., 2022),⁴ a public dataset created by scraping images and their captions from the web. We focus on two versions, Stable Diffusion 1.4 and 1.5, which are both trained on text-image pairs from the 2.3 billion English portion of LAION-5B.⁵

2.2 Gender Classification

We analyze bias in images with respect to *perceived* gender.⁶ To classify binary gender at scale, we utilize an automatic classifier, which we validate with a human study (see Appendix

²We define bias as a deviation from the 50% balanced (binary) gender ratio. This definition differs from measuring performance gaps between groups (e.g., TPR difference), which is common in classification setups.

³Since we study gender bias, prompts exclude explicit gender information to avoid skewing generations.

⁴LAION was publicly available at the time of writing this paper, but has since been removed.

⁵Stable Diffusion 1.5 is finetuned for a longer duration on LAION-Aesthetics (a subset of higher quality images).

⁶Classifying binary gender based on appearance has limitations and perpetuates stereotypes, and excludes non-binary gender identities.

A.4). It is important to verify that images include faces, and that perceived gender is discernible from these images. Therefore, we first check whether an image contains a single face using a face detector.⁷ Then, we use CLIP (Radford et al., 2021), a multimodal model with zero-shot image classification capabilities, to predict gender (note that Stable Diffusion also uses CLIP’s text encoder to encode prompts). We perform gender classification by computing the cosine similarity between CLIP embeddings: $\cos(\text{CLIP}(\text{image}), \text{CLIP}(\text{a photo of a woman}))$ as well as $\cos(\text{CLIP}(\text{image}), \text{CLIP}(\text{a photo of a man}))$. After computing image-text similarities, we normalize the similarity values using softmax to obtain gender probabilities. We follow the zero-shot procedure described in Radford et al. (2021) and use the texts “a photo of a woman” and “a photo of a man” for computing image-text similarities.⁸

To exclude cases where gender is difficult to infer (e.g., faces might be blurred or obscured), we only consider images for which the predicted gender probability is greater than or equal to 0.9. We apply these filtering steps to both training and generated images.

2.3 Occupations

Similarly to previous works, we analyze gender-occupation biases for occupations with varying levels of bias (Rudinger et al., 2018; Zhao et al., 2018; De-Arteaga et al., 2019). These include occupations that skew male (e.g., CEO, engineer), fairly balanced (e.g., attorney, journalist), and female (e.g., dietitian, receptionist) based on the training data. In total, we consider 62 job occupations, which can be found in Table 4 in the Appendix.

3 Methodology

3.1 Measuring Model Bias

To measure biases exhibited by the model, we generate images using four prompts, shown in Table 1. These prompts deliberately do not contain gender information since we want to capture biases learned by the model. Both prompts #1 and #2 also direct the model to generate faces by including “face” and “portrait”, respectively. We generate 500 images per occupation and prompt using various random

⁷https://developers.google.com/mediapipe/solutions/vision/face_detector/python.

⁸Applying CLIP to infer gender on the FairFace dataset (Karkkainen and Joo, 2021) results in strong performance (> 95% accuracy) across various racial subgroups.

#	Prompt
1	A photo of the face of a/an [OCCUPATION]
2	A portrait photo of a/an [OCCUPATION]
3	A photo of a/an [OCCUPATION] smiling
4	A photo of a/an [OCCUPATION] at work

Table 1: The four prompts we use to generate images. “[OCCUPATION]” is a placeholder we replace with one of the 62 occupations we use (e.g., engineer).

seeds to initialize random noise. We define G_{P_o} as the percentage of females in generated images for a prompt P describing an occupation o .

3.2 Measuring Data Bias

Given that the training data consists of image-caption pairs, we use captions to obtain relevant training examples. In doing so, we assume that the captions relating to a given occupation mention the occupation. We use the search capabilities of WIMBD (Elazar et al., 2024), a tool that enables exploration of large text corpora, to query LAION. We define T_{S_o} as the percentage of females in images for a training subset S corresponding to occupation o (we provide more details on how training examples are selected in Section 4).

3.3 Evaluating Bias Amplification

We compute bias amplification by comparing the female percentage in generated images (G_{P_o}) vs. training images (T_{S_o}) for a specific occupation o using the approach outlined in Zhao et al. (2017):

$$A_{P_o, S_o} = |G_{P_o} - 50| - |T_{S_o} - 50|$$

This formulation takes into account that amplification for a given occupation is specific to the prompt P_o used to generate images, as well as the chosen subset of training examples S_o . For a set of occupations O , the expected amplification is:

$$\mathbb{E}_{o \in O} [A_{P_o, S_o}] = \frac{1}{|O|} \sum_{o \in O} A_{P_o, S_o}$$

A_{P_o, S_o} is calculated for each occupation and is aggregated across occupations (O) to obtain $\mathbb{E}[A_{P_o, S_o}]$ for each prompt. We then average $\mathbb{E}[A_{P_o, S_o}]$ across all four prompts. For occupations that skew male in the training data, bias is amplified if it skews further male in generated images, and vice versa for occupations that skew female. Bias decreasing from training to generation is considered de-amplification. We exclude occupations that exhibit different directions of bias at training

Example Captions
Portrait of smiling young female mechanic inspecting a CV joint on a car in an auto repair shop
Young male nurse wearing surgical antiviral mask
Muscular bearded athlete drinks water after good workout session in city park
Portrait of a salesperson standing in front of electrical wire spool with arms crossed in hardware store
Radiology technician performing mammography scan

Table 2: Training captions often include additional details (e.g., descriptions, actions) that reduce ambiguity, and may contain explicit and implicit gender information. In contrast, the prompts we use to generate images (Table 1) lack context and specificity.

and generation from our analysis since they do not adhere to our definition of amplification/de-amplification (i.e. model bias is more/less severe than training bias *along the same direction*).

4 Estimating Bias Amplification: A Baseline

We examine the extent to which Stable Diffusion amplifies gender-occupation biases from the data by selecting training examples that contain a given occupation in the caption (e.g., all captions that contain the word “president”). In practice, we randomly sample a subset of 500 training examples containing each occupation. We find that Stable Diffusion amplifies bias relative to the training data by 12.57%⁹ on average across all occupations and prompts (10.24% for Prompt #1, as shown in Figure 2). This behavior is concerning because instead of reflecting the training data and its statistics, the model compounds bias by further underrepresenting groups. However, when qualitatively inspecting examples, we observe discrepancies in how occupations are presented in captions vs. prompts due to varying levels of ambiguity.

For example, we notice the use of explicit *gender indicators* to emphasize deviations from stereotypical gender-occupation associations, such as female engineers in Figure 1. While gender information is used frequently in captions, we hypothesize that usage is more common for underrepresented groups. If this hypothesis holds, the gender distribution would shift closer towards balanced in resulting training images. As a result, the decision to focus

⁹We report values for Stable Diffusion 1.4 throughout the paper, but results for both model versions are presented in Table 3. Overall, we observe similar trends for both models.

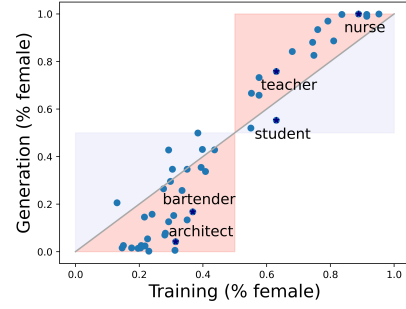


Figure 2: **Bias is amplified consistently using our baseline approach (by 10.24% for Prompt #1).** The x-axis corresponds to the % female in training images, and the y-axis corresponds to the % female in generated images (using Prompt #1). Each point represents an occupation. Shading: **Amplification** and **De-Amplification**.

on all captions vs. captions without any gender indicators can exaggerate amplification measures, as we see in Section 5.1.

More generally, prompts commonly used to study gender-occupation bias are intentionally underspecified, or lack detail. Underspecification results in the model having to generate images from textual inputs that are vague and open to interpretation (Hutchinson et al., 2022; Mehrabi et al., 2023). For instance, the prompt “A photo of the face of a/an [OCCUPATION]” does not contain any adjectives or information about surroundings, activities, etc. In contrast, captions from the training data may contain context and details that result in less ambiguous descriptions, as shown in Table 2.¹⁰

Discrepancies in how captions and prompts are written also impact how occupations are depicted in training and generated images. These differences are especially notable for occupations that have multiple interpretations. For example, when querying for training examples containing “president”, the resulting captions may refer to various types of presidents, including the president of a company or organization, as shown in Figure 3a. However, when generating images using the prompt “A photo of the face of a president”, the model appears to interpret president as a leader of a country, often the United States (we also showcase similar differences for the occupation teacher in Figure 3b). Given that there are evident qualitative differences in images, we should not expect the training and generation gender distributions to match.

To compare bias at training and generation, we need to consider gender ratios for similar captions and prompts. Therefore, we cannot conclude

¹⁰We showcase examples that include descriptions of individuals and activities they are engaged in.



(a) Training captions for **President**: 1) “Leana Wen, Planned Parenthood president...” 2) “New Schaumburg Business Association President...” 3) “BCCI president N Srinivasan...” 4) “Indiana Pacers president of basketball operations...”

(b) Training captions for **Teacher**: 1) “Brad Draper, percussion teacher...” 2) “teacher/author in the 80s sits in yoga lotus pose...” 3) “Jo Anne Young Art Teacher...” 4) “Classical Guitar Teacher...”

Figure 3: **Differences between training and generated examples using our baseline approach.** Here, we handpick examples of discrepancies in how occupations are depicted in training vs. generated examples for **President** (left) and **Teacher** (right) professions. For instance, the model interprets “president” as a president of a country, often the U.S., while the term president can refer more broadly to a president of a company or organization, as shown in the shortened captions. Border shading: perceived gender is **female** and **male**.

whether differences in gender ratios are due solely to the model amplifying bias, or other confounding factors that contribute to amplification. Next, we focus on decreasing the impact of distribution shifts on bias amplification evaluation.

5 Estimating Bias Amplification: Reducing Distributional Differences

In this section, we reduce training and generation discrepancies by restricting the search space of training examples. The prompts P_o remain fixed, while the subset of training examples S_o varies.

5.1 Excluding Explicit Gender Indicators

A notable distinction between training and generation is the use of explicit gender indicators, which is absent from the prompts we use. On average, more than half the captions (59.5%) contain explicit gender information. It is important to note that gender indicators can emphasize the underrepresented gender for a given occupation. For example, images of female mechanics in the training data frequently accompany captions that indicate the mechanic is female. In comparison, this specification is less common for male mechanics (only 30% of male mechanic examples contain explicit gender indicators, as opposed to 68% for female mechanics, using the approach discussed in Appendix A.5).

To validate these observations, we compute the correlation between the percentage of females in training images and the percentage of corresponding captions with female indicators. We expect that female-skewing occupations are less likely to con-

tain explicit female gender indicators in captions, resulting in a negative correlation. The Pearson’s correlation coefficient is indeed negative, with a coefficient value of -0.458 and statistically significant (significance level < 0.05). These results suggest that including training examples with gender mentions in evaluations may exaggerate amplification.

Addressing Gender Indicators To assess whether amplification differs for the subset of captions without indicators, we split the training examples selected in Section 4 by detecting explicit gender mentions in the captions (more details in Appendix A.5 and example image-caption pairs in Table 5). We focus on the subset of captions, S_o , without explicit male or female indicators.

Reduced Bias Amplification We observe that bias amplification is noticeably lower when focusing on the no-gender indicator subset of training examples. Compared to the initial amplification of 12.57% for our baseline, the average amplification for captions without gender indicators is 8.66% ($\downarrow 31\%$), as shown in Table 3. This behavior aligns with the reasoning described above — gender indicators often delineate the presence of the underrepresented gender, which in turn inflates amplification measures.

5.2 Nearest Neighbor Captions

Beyond explicit gender indicators, there are clear differences in the information conveyed by prompts vs. captions. The prompts we use are concise and structured, but lack concrete details. On the other

hand, randomly sampled training captions are more diverse and vary in their usage of the occupation and contextual information, as highlighted in Table 2 and Figure 3. Furthermore, captions may contain implicit gender information (e.g., descriptors, attire, activities) that is absent from prompts.

These qualitative differences are also apparent when comparing caption and prompt text embeddings. We use SBERT (Reimers and Gurevych, 2019) to compute text embeddings,¹¹ and calculate the average pairwise cosine similarity between caption and prompt embeddings for each occupation. We find that the average cosine similarity across occupations is 0.385, indicating that caption and prompt similarity is relatively low (relative to nearest neighbors, which we will see next).

Addressing Similarity Discrepancies To account for these gaps, we propose using nearest neighbors (NN) to select captions that closely resemble prompts. We can find neighbors by considering all captions that contain a given occupation, and selecting examples based on the similarity between caption and prompt text embeddings instead of sampling randomly. As a result, the chosen captions are closer in structure and wording to prompts. We compute the cosine similarity between text embeddings to measure the similarity between captions and prompts.¹² For a given occupation, we consider the top- k similar captions, where $k = 500$.

Applying NN, the average cosine similarity between caption and prompt embeddings increases to 0.704 ($\uparrow 83\%$ from the naive approach), which occurs by design since we directly target examples that resemble prompts. Note however, that the increase in similarity is also reflected in image embeddings. The pairwise similarity of CLIP image embeddings increases with NN ($\uparrow 13\%$ from the naive approach), indicating that chosen training and generated images are slightly more similar.

There are noticeable qualitative improvements as well. NN chooses captions that are closer in structure and meaning to prompts (e.g., “Picture of a teacher in the classroom”), which also impacts corresponding training images. In contrast to the naive approach, the training images corresponding

¹¹Specifically, we use the all-MiniLM-L6-v2 model to compute text embeddings with SBERT.

¹²Text embeddings used to compute NN can reinforce biases. By using SBERT (Reimers and Gurevych, 2019), we avoid leaking biases from Stable Diffusion’s text encoder (CLIP) for selecting training examples.



(a) Training captions for **President**: 1) “The president is smiling.” 2) “President Donald J. Trump - Official Portrait of President George H. W. Bush” 3) “Portrait of President Ronald Reagan” 4) “Official Portrait of President Ronald Reagan”
(b) Training captions for **Teacher**: 1) “Picture of a teacher in the classroom.” 2) “Portrait of a smiling teacher in a classroom.” 3) “Portrait of teacher woman working” 4) “Teacher smiling in classroom, portrait”

Figure 4: **Training examples chosen with Nearest Neighbors** (using the prompt “A photo of the face of a/an [OCCUPATION]”). Selected training captions and images are more similar to prompts and generated images as compared to the examples in Figure 3. Border shading: perceived gender is **female** and **male**.

to NN captions for “president” primarily represent world leaders (often US presidents), while captions for “teacher” depict educators in classroom settings, as shown in Figure 4.

Reduced Bias Amplification When selecting training examples S_o using NN, we see that bias amplification reduces considerably across occupations and prompts, as shown in Table 3. The average amplification drops to 6.76% ($\downarrow 46\%$ relative to the naive approach). While NN yields increased similarity between training and generated examples, there are still unresolved sources of distribution shift that impact amplification measures.

5.3 Combining Approaches

We observe that amplification further reduces when combining the no-gender indicator subset with NN, as shown in the last rows in Table 3. The average amplification decreases to 4.35% ($\downarrow 65\%$ from the naive approach), which is noticeably lower compared to the values for each method individually. Both methods work in tandem to reduce distributional differences in complementary ways, perhaps by targeting both explicit and implicit gender information. We also observe greater reductions for specific prompts (e.g., amplification is just 1.11% for Prompt #1), which indicates that distribution shifts are more effectively addressed for some prompts than others.

We perform a one-sample t-test to test the null

Approach	SD 1.4					SD 1.5				
	#1	#2	#3	#4	Average	#1	#2	#3	#4	Average
Naive Approach	10.24	17.57	10.77	11.68	12.57	10.87	16.36	11.15	9.91	12.07
No Gender Indicators	6.49	13.58	7.09	7.49	8.66	6.76	12.41	6.82	5.87	7.97
Nearest Neighbors (NN)	3.59	12.62	5.58	5.27	6.76	4.01	11.14	5.21	3.65	6.01
NN + No Indicators	1.11	8.72	3.06	4.05	4.35	1.55	7.29	2.78	2.72	3.59

Table 3: **Bias Amplification across occupations using Stable Diffusion (SD) 1.4 and 1.5.** Results are shown for each prompt and averaged across prompts. Amplification lowers considerably when excluding captions with gender indicators and using nearest neighbors to select captions. We see further reductions when combining approaches.

hypothesis that the expected amplification is 0 for each of the prompts; we fail to reject the null hypothesis for prompts #1 and #3 and reject the null hypothesis for prompts #2 and #4 (significance level < 0.05). Our results indicate a portion of amplification is unexplained for all prompts, especially prompts #2 and #4, and may involve more subtle confounding factors. Although the proposed methods do not account entirely for discrepancies between training and generation, we observe that the bias measures become closer as we select subsets of training captions that resemble prompts.

6 Removing Distributional Differences: A Lower Bound

The previous approaches reduce discrepancies between training and generation by evaluating amplification with captions that are more similar to prompts. Instead, we can focus our efforts in the other direction and modify the prompts we use to align with captions more closely. One way to achieve this is to eliminate differences altogether by making prompts and captions identical. We then ask: *Does using identical texts to measure training and generation bias lower amplification?* We use the original training subset (S_o) from Section 4 and make the prompts (P_o) match the captions verbatim. In this setup, we generate 10 images for every prompt in P_o , and then compute amplification using $P_o := S_o$ for each occupation.

We hypothesize that enforcing prompts and captions to match yields similar bias measurements, which reduces amplification. As shown in Figure 5a, amplification is small when $P_o = S_o$ and most occupations reside along the diagonal (no amplification). The average amplification drops to 0.68%, indicating that the model mostly reflects training bias.¹³ Furthermore, amplification remains consistently low, even for highly imbalanced occupations.

¹³However, we reject the null hypothesis that the expected amplification is 0 using a one-sample t-test.

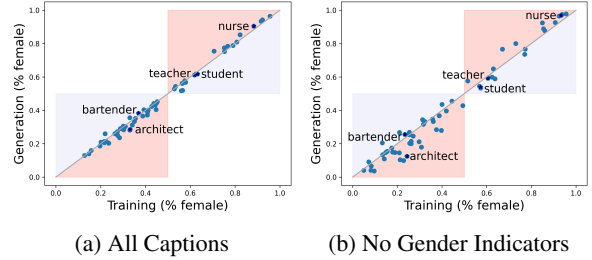


Figure 5: **Bias amplification when prompting with training captions.** We observe minimal amplification when $P_o = S_o$ (left). This behavior mostly holds when focusing on captions without explicit gender indicators (right). Shading: **Amplification** and **De-Amplification**.

For captions that contain either male or female gender indicators, the model generates images that match the gender of corresponding training images (with 98.41% accuracy), since this information is directly provided in the prompt. Therefore, we analyze the results separately on the subset of captions without gender indicators. As shown in Figure 5b, bias amplification is larger for the no gender indicator subset as compared to all captions. That being said, the average amplification remains low at 2.05% ($\downarrow$ 84% relative to our naive approach).¹³ We also observe similar results when paraphrasing the original training captions and using these texts as prompts, as discussed in Appendix A.6, which suggests our results are not simply due to memorization of captions.

Although practitioners are unlikely to utilize prompts that exactly match training captions, this experiment highlights the impact of distributional similarity between captions and prompts when comparing biases. In addition, it provides a lower bound to the bias amplification problem. In summary, we conclude that the model nearly mimics biases from the data when we eliminate distributional differences.

7 Related Work

Relating pretraining data to model behavior
There is a growing body of work focused on study-

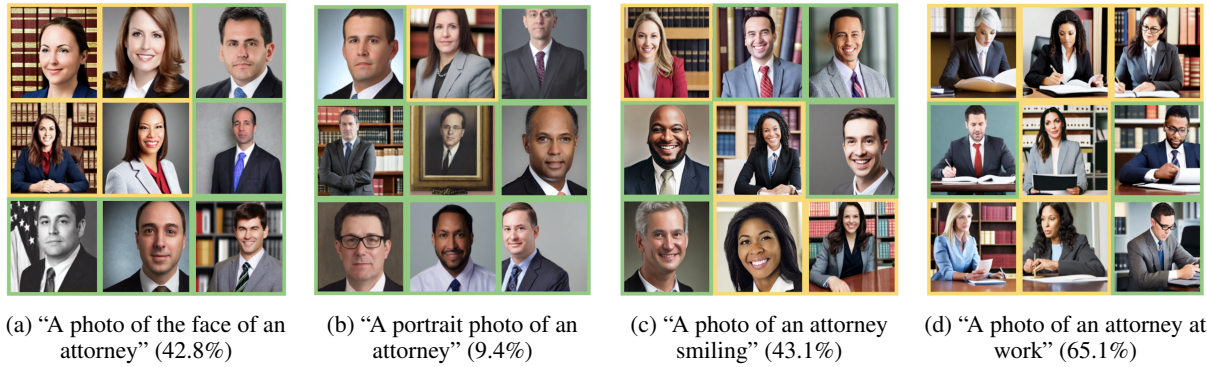


Figure 6: **Generations for “attorney” using different prompts.** Specific wording choices in prompts lead to notable differences in the percentage of generated images that are predicted as female. Border shading: perceived gender is **female** and **male**.

ing pretraining data properties and their relation to model behavior. This type of large-scale data and model analysis provides useful insights into model learning and generalization capabilities (Carlini et al., 2023). Recent work shows that few-shot capabilities of large language models are highly correlated with pretraining term frequencies, and that models struggle to learn long-tail knowledge (Kandpal et al., 2023; Razeghi et al., 2022). Several works have also explored the relationship between pretraining data and model performance from a causal perspective (Biderman et al., 2023; Elazar et al., 2022; Longpre et al., 2023). For example, Longpre et al. (2023) comprehensively investigate how various data curation choices and pretraining data slices affect downstream task performance.

Bias Amplification Our work is strongly inspired by the findings of Zhao et al. (2017), who show that structured prediction models amplify biases present in the data. However, there are important differences to note. First, their task jointly predicts multiple target labels (including gender), as opposed to generating images. Furthermore, their work largely focuses on mitigating amplification, as opposed to investigating underlying factors that affect amplification. Hall et al. (2022) consider how data, training, and modeling choices influence amplification using a classification setup with synthetic bias, but do not examine distribution shifts.

Friedrich et al. (2023) also compare biases exhibited by LAION and Stable Diffusion, and show that the model demonstrates amplification. Instead of identifying relevant training examples using captions, they use image-text similarity between training images and prompts. Furthermore, their work primarily focuses on bias mitigation, while our work is centered around analyzing confounding

factors that impact amplification.

Bias in text-to-image models While it is well-established that language and vision models are prone to biases individually, recent work has shown that text-to-image models display similar biases. Several works analyze various biases in text-to-image models, including gender biases (Wu et al., 2023; Zhang et al., 2023), geographical disparities (Basu et al., 2023; Naik and Nushi, 2023), and intersectional biases (Fraser et al., 2023; Luccioni et al., 2023). Bianchi et al. (2023) demonstrate that stereotypes persist even after using counter-stereotypes. However, these works solely evaluate model biases, and do not examine the training data.

8 Discussion

Our results bring up a number of key issues.

Generalizability Our work demonstrates that using naive procedures to evaluate bias amplification can lead to exaggerated amplification measures. While our analysis does not account for all sources of distribution shift that contribute to amplification, it is meant to be illustrative. We encourage future studies to build on our findings by examining different experimental setups (i.e., datasets, models, and types of bias) to gain a more comprehensive understanding of bias amplification and the impact of confounding factors.

Variation Across Prompts As we highlight in Figure 6, small changes to prompts can have a resonating effect on conclusions about bias. For example, “A portrait photo of an attorney” skews heavily male while “A photo of an attorney at work” skews female in generated images. Furthermore, reductions in amplification differ based on the prompt (e.g., 89% reduction for Prompt #1 vs. 49% for

Prompt 2), indicating that there are prompt-specific sources of distribution shift. Our findings match prior work demonstrating that model biases are highly sensitive to wording and phrasing choices (Seshadri et al., 2022; Selvam et al., 2023).

Amplification Baseline Our interpretation of amplification is centered around models exacerbating biases in the training data as opposed to real-world statistics (Kirk et al., 2021; Bianchi et al., 2023). Both approaches are useful to study but answer fundamentally different questions. Our approach offers insights into whether model behavior reflects the training data, while real-world amplification captures how well the model reflects reality.

Connection to Simpson’s Paradox The title of our paper alludes to Simpson’s Paradox (Simpson, 1951), a phenomenon in which a trend or relationship observed in subgroups within the data reverses or disappears when subgroups are combined. We draw direct parallels to our analysis and insights; although we observe substantial amplification in our initial setup, amplification reduces drastically after selecting specific subsets of the training data and decreasing the impact of confounding factors.

Recommendations Our findings underscore how distribution shifts contribute to bias amplification, which has important implications. Those involved in data-focused efforts should consider how practitioners specify prompts and interact with models when curating training data. Alternatively, crowdsourcing or automatically rewriting existing training captions to reflect real-world model usage may result in lower amplification. Additionally, we recommend that evaluations use multiple prompts and remove prompt-specific confounding factors (e.g., by using NN to select relevant training examples).

9 Conclusion

In summary, we investigate whether Stable Diffusion amplifies gender-occupation biases by comparing training data and model biases. We highlight how naive evaluations of amplification fail to consider distributional differences between training and generation, which leads to a misleading understanding of model behavior. Although amplification is not eliminated entirely, we observe that reducing discrepancies between captions and prompts during evaluation results in substantially lower measurements. We recommend that analyses comparing training data and model biases, or any

data and model properties more generally, account for various distribution shifts that skew evaluations.

Limitations

Beyond the training data, another source of bias is the text embeddings obtained from CLIP. By solely comparing biases in the data vs. those exhibited by Stable Diffusion, our analysis overlooks biases that arise from encoding prompts. As a result, we cannot disentangle how much this component impacts overall amplification. Note that the effect of such an external embedding cannot be easily accounted for, since CLIP’s training data is not public. More work is needed to understand the impact of using external, frozen models as a model component.

Additionally, we automate gender classification using CLIP because previous works have shown that CLIP gender predictions align with human annotations and CLIP gender classification performance on the FairFace dataset¹⁴ is strong ($> 95\%$) across various racial categories. Nevertheless, we recognize the limitations of using a model to classify gender in images, since CLIP inherits biases from its training data.

Ethics Statement

Scope of Work Our work centers around critically examining bias amplification evaluation. The approaches we propose to reduce distribution shifts observed during evaluation do not solve underlying gaps between the data used to train models and how users interact with models. Rather, they serve to deepen our understanding of why models amplify biases present in the training data. Ideally, our findings will motivate future work on 1) thorough and nuanced evaluations of bias amplification and 2) fundamentally addressing training and generation discrepancies from a data perspective.

Bias Definition Our work focuses on a narrow slice of social bias analysis by studying gender-occupation stereotypes. Since models exhibit various types of discriminatory bias (e.g., racial, age, geographical, socioeconomic, disability, etc.), as well as intersectional biases, it is equally important to perform evaluations for these definitions of bias. Furthermore, we only consider binary gender, which has clear drawbacks. Our analysis ignores how text-to-image models perpetuate biases

Our code is released here: <https://github.com/preethisheshadri518/bias-amplification-paradox>

¹⁴<https://github.com/joojs/fairface>

for non-binary identities and relies on information such as appearance and facial features to infer gender in training and generated images, which can propagate gender stereotypes.

Geographical Diversity The captions and prompts used to study bias are solely written in English. We hope future work will shed light on multilingual bias amplification in text-to-image models. It is also worth noting that the gender-guesser library (infers gender from names) likely performs worse on non-Western names. The documentation mentions that the library supports over 40,000 names and covers a “vast majority of first names in all European countries and in some overseas countries (e.g., China, India, Japan, USA)”. Therefore, the name coverage (or lack thereof) impacts our ability to identify captions with gender information.

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A Appendix

A.1 Occupations

A full list of occupations is shown in Table 4. We exclude occupations that switch bias directions in our analysis since they do not adhere to our definition of amplification/de-amplification (i.e. model bias is more/less severe than training bias *along the same direction*). The extent to which these swaps occur depends on the prompt and model version, since model bias results are specific to both. More research is needed to understand and explain this behavior.

There are 9 occupations that exhibit switching behavior consistently for all prompts using SD 1.4 and 6 occupations using SD 1.5 (5 occupations are common to both). In many of these cases, there are strong deviations from 50% at training, generation, or both. For example, in the case of “painter”, even though the training bias is 52.6%, the average bias is 17.5% for SD 1.4 and 19.3% for SD 1.5.

Tables 6 (SD 1.4) and 7 (SD 1.5) show bias values for each occupation at both training and generation. For some occupations (e.g., attorney, cook, surgeon), the gender distributions in generated images varies considerably across prompts.

A.2 LAION

LAION is a large dataset of image-caption pairs released under CC-BY 4.0. Instead of saving scraped images, LAION stores URLs that link to the images, which we then use to download images. We only download a subset of examples that pertain to the occupations in Table 4.

There are notable issues to point out with LAION. Since the dataset only contains URLs to images, some of these URLs may suffer from link rot and may no longer be accessible. Therefore, it is impossible to reproduce the exact set of images used during training. Furthermore, the dataset includes copyrighted and not-safe-for-work (NSFW) content. We acknowledge these issues and emphasize that our use of LAION is for research purposes to 1) analyze gender-occupation biases in the data and 2) evaluate bias amplification.

A.3 Generating Images

Stable Diffusion 1.4 and 1.5 contain roughly 1 billion parameters. Using a single TITAN RTX GPU, it takes about 3.5 seconds to generate one image. To generate 500 images for each occupation ($\times 62$),

prompt ($\times 4$), and model version ($\times 2$), it takes approximately 240 hours. We use the default generation parameters, which include a guidance scale of 7.5 and 50 inference steps. We also use a batch size of 4.

A.4 Image Gender Classification

While CLIP is susceptible to biases (Hall et al., 2023), its gender predictions have been shown to align with human-annotated gender labels (Bansal et al., 2022; Cho et al., 2023). In addition, we perform human evaluation with 7 participants on 200 randomly selected training and generated images. We ask participants to provide binary gender annotations (or indicate that they are unsure), and find that Krippendorff’s coefficient, which measures inter-annotator agreement, is high ($\alpha = 0.948$). Additionally, 98% of CLIP predictions match the majority vote annotations.

A.5 Explicit Gender Indicators

To identify captions with explicit gender information, we consider 1) gender words (male, female, man, woman, gent, gentleman, lady, boy, girl), 2) binary gender pronouns (he, him, his, himself, she, her, hers, herself), and 3) names. We perform named entity recognition using the *en_core_web_lg* model from spaCy to identify name mentions, and then use the gender-guesser library <https://pypi.org/project/gender-guesser/> to infer gender. We include example training captions with explicit gender mentions in Table 5. After excluding examples with gender indicators (§5.1), the average number of examples remaining is 202 out of 500 training examples (40.4%) per occupation.

A.6 Paraphrasing Captions

In Section 6, we align the train and test distributions by directly prompting the model with training captions. We show that amplification is minimal when eliminating distributional differences. As a follow-up, we study what happens if we instead use prompts that are similar but not identical to training captions when evaluating amplification. To construct similar examples, we paraphrase the original captions using gpt-3.5-turbo. We set the temperature to 0 and use the following prompt to generate paraphrases:

*Please paraphrase the phrase/sentence below.
You can change words without changing the*

original meaning or intent. You must include the word [OCCUPATION].
Phrase/Sentence: [CAPTION]

Using the training subset S_o from Section 6 and the paraphrased captions as prompts P_o , we find that amplification still remains low — amplification is 0.69% for all captions (compared to 0.68% in Section 6) and 2.49% for captions without explicit gender indicators (compared to 2.05% in Section 6). The paraphrasing results indicate that our original findings from Section 6 extend beyond using exact training captions. In other words, these results suggest the model can generalize, and does not rely solely on memorization to achieve low amplification.

Occupations				
accountant	dentist	journalist	poet	singer
architect	dietitian	lawyer	politician	student
assistant	doctor	librarian	president	supervisor
athlete	engineer	manager	prime minister	surgeon
attorney	entrepreneur	mechanic	professor	teacher
author	fashion designer	musician	programmer	technician
baker	filmmaker	nurse	psychologist	therapist
bartender	firefighter	nutritionist	receptionist	tutor
ceo	graphic designer	painter	reporter	veterinarian
chef	hairdresser	pharmacist	researcher	writer
comedian	housekeeper	photographer	salesperson	
cook	intern	physician	scientist	
dancer	janitor	pilot	senator	

Table 4: List of 62 occupations used to study gender-occupation biases.

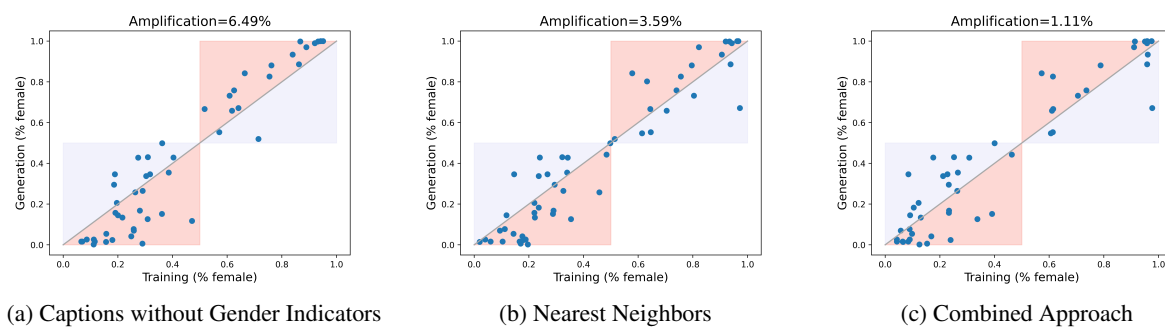


Figure 7: **Bias amplification for various approaches to address discrepancies between training and generation.** The proposed approaches yield lower bias amplification, especially the combined method (c). Results are shown for Prompt #1 specifically. Regions are shaded based on **Amplification** and **De-Amplification**.

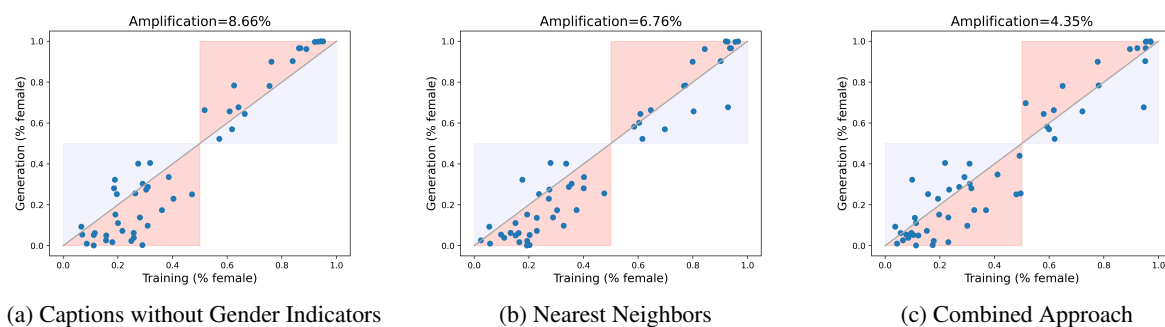


Figure 8: **Bias amplification for various approaches to address discrepancies between training and generation.** The proposed approaches yield lower bias amplification, especially the combined method (c). Results are averaged across all prompts. Regions are shaded based on **Amplification** and **De-Amplification**.

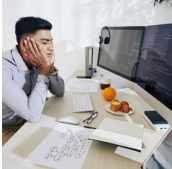
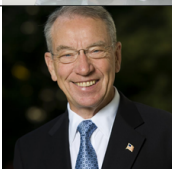
Image	Caption	Gender Indicator
	Portrait of young woman programmer working at a computer in the data center filled with display screens	woman
	Tired young indian programmer almost sleeping at his desk after working on difficult project all day long	his
	Female accountant very busy in office	female
	Accountant managing manual bill monitoring tasks in his home office	his
	Iowa Republican Senator Chuck Grassley	first name
	U.S. Senator Kirsten Gillibrand (D-NY) pauses during a news conference on Capitol Hill in Washington	first name
	Portrait of young male mechanic in bicycle store, Beijing	male
	African american woman mechanic repairing a motorcycle in a workshop	woman
	Attractive woman photographer taking images with dslr camera outdoors in park.	woman
	Photographer John G. Zimmerman with his pipe and Hucher camera, 1972.	first name/his

Table 5: Example training images and captions with explicit gender indicators for select occupations (in bold).

Occupation	Training	Prompt #1	Prompt #2	Prompt #3	Prompt #4
accountant	29.8	29.5	3.4	43.8	35.7
architect	31.4	4.2	2.2	3.0	0.0
assistant	44.6	67.1	56.3	71.9	75.6
athlete	44.8	80.0	51.9	69.3	77.3
attorney	29.2	42.8	9.4	43.1	65.1
author	42.8	83.6	53.0	81.5	61.0
baker	41.4	81.1	31.2	58.8	59.3
bartender	36.8	16.8	2.6	12.9	22.9
ceo	15.0	2.6	1.8	4.8	11.9
chef	28.0	7.0	1.2	1.4	5.8
comedian	21.8	2.4	0.0	3.6	1.0
cook	35.0	34.7	8.6	49.4	69.3
dancer	81.0	88.7	98.8	99.0	100.0
dentist	58.6	41.4	4.4	29.2	41.8
dietitian	95.2	100.0	100.0	100.0	99.8
doctor	40.8	33.7	3.8	14.6	57.6
engineer	20.6	2.6	0.2	1.2	0.0
entrepreneur	43.6	42.8	1.8	12.8	34.6
fashion_designer	76.0	93.4	80.8	89.8	97.2
filmmaker	29.2	12.6	3.2	8.3	14.9
firefighter	14.6	1.6	1.0	15.9	3.2
graphic_designer	52.8	11.8	14.4	32.7	41.6
hairstylist	79.2	97.0	95.6	94.6	97.6
housekeeper	91.4	99.0	99.8	100.0	100.0
intern	57.6	65.8	31.5	77.2	53.4
janitor	20.4	1.6	3.0	14.6	5.7
journalist	38.4	49.9	59.9	68.8	64.0
lawyer	27.6	26.5	8.0	39.0	47.7
librarian	74.4	88.1	83.6	93.6	94.8
manager	13.0	20.6	7.8	29.7	42.8
mechanic	17.6	1.6	0.0	0.2	35.3
musician	22.6	5.4	4.2	7.2	3.2
nurse	88.8	100.0	100.0	100.0	100.0
nutritionist	83.6	99.8	92.8	96.6	97.5
painter	52.6	36.4	12.2	17.6	3.6
pharmacist	68.0	84.2	26.9	54.9	91.7
photographer	55.0	52.0	27.5	46.5	13.2
physician	39.4	35.5	2.0	37.5	59.3
pilot	30.4	34.7	12.2	66.3	15.9
poet	30.8	15.2	2.0	19.5	32.8
politician	21.6	14.5	4.2	15.9	9.6
president	19.6	1.4	0.2	8.0	0.8
prime_minister	24.0	15.7	10.6	13.2	21.4
professor	28.2	7.8	2.8	9.2	5.3
programmer	23.0	0.2	0.0	0.2	0.0
psychologist	58.6	44.3	21.6	57.2	52.9
receptionist	91.4	99.8	100.0	99.8	99.8
reporter	44.4	54.8	55.2	55.1	67.8
researcher	44.6	80.2	41.8	67.6	50.9
salesperson	39.8	43.0	5.2	33.1	33.7
scientist	33.4	25.7	24.0	29.3	23.2
senator	35.0	13.4	2.0	8.2	5.4
singer	57.6	73.2	60.3	69.2	60.1
student	63.0	55.3	48.5	62.1	43.3
supervisor	65.2	18.3	4.8	16.6	14.9
surgeon	30.2	82.5	15.6	67.6	82.5
teacher	63.0	75.8	55.7	94.0	88.0
technician	31.2	0.6	0.0	0.6	0.0
therapist	74.8	82.6	63.3	79.2	87.5
tutor	59.2	48.1	23.1	32.7	43.5
veterinarian	55.2	66.7	44.7	64.1	89.9
writer	30.2	73.3	30.1	76.0	63.8

Table 6: The percentage of females across occupations in training images (using our initial approach from Section 4) and generated images using **SD 1.4**. We display generation results for each prompt. **Highlighted rows** indicate occupations for which bias switches direction from training to generation across all prompts.

Occupation	Training	Prompt #1	Prompt #2	Prompt #3	Prompt #4
accountant	29.8	34.9	5.4	42.1	45.2
architect	31.4	10.0	2.2	2.2	3.4
assistant	44.6	69.2	60.8	58.6	77.8
athlete	44.8	76.6	46.0	50.0	74.3
attorney	29.2	50.8	11.7	44.3	68.3
author	42.8	88.2	57.4	75.4	69.0
baker	41.4	82.3	33.9	53.3	66.6
bartender	36.8	10.0	2.2	4.8	12.2
ceo	15.0	1.4	2.0	5.4	18.5
chef	28.0	12.0	0.8	1.4	7.0
comedian	21.8	1.6	0.0	1.4	0.6
cook	35.0	38.4	16.4	43.5	75.1
dancer	81.0	83.8	97.4	97.6	100.0
dentist	58.6	41.9	5.4	22.7	20.4
dietitian	95.2	100.0	100.0	100.0	99.8
doctor	40.8	38.2	8.8	12.6	53.4
engineer	20.6	10.6	0.6	1.6	0.0
entrepreneur	43.6	59.7	4.6	16.9	41.6
fashion_designer	76.0	97.4	90.3	92.2	98.6
filmmaker	29.2	18.4	5.2	8.8	7.8
firefighter	14.6	1.4	0.2	12.5	4.5
graphic_designer	52.8	22.6	15.3	29.5	63.3
hairstylist	79.2	99.6	98.0	95.4	97.3
housekeeper	91.4	99.6	100.0	100.0	100.0
intern	57.6	72.6	37.1	68.8	60.4
janitor	20.4	3.6	3.2	8.4	6.2
journalist	38.4	57.2	60.2	59.7	60.7
lawyer	27.6	34.1	8.8	36.8	48.2
librarian	74.4	93.4	85.8	87.8	94.6
manager	13.0	24.0	14.2	28.7	41.3
mechanic	17.6	6.4	0.2	1.0	20.8
musician	22.6	5.4	1.4	2.8	2.8
nurse	88.8	100.0	100.0	100.0	100.0
nutritionist	83.6	99.8	97.8	97.2	98.0
painter	52.6	43.7	20.0	10.6	2.7
pharmacist	68.0	87.3	26.1	49.6	83.8
photographer	55.0	58.1	32.5	44.8	26.0
physician	39.4	46.4	3.2	36.5	62.0
pilot	30.4	20.9	11.4	35.3	7.5
poet	30.8	12.4	2.6	11.6	42.1
politician	21.6	24.9	10.2	16.7	15.7
president	19.6	4.6	0.4	12.9	2.2
prime_minister	24.0	25.5	23.0	20.0	42.9
professor	28.2	9.2	3.0	5.6	8.6
programmer	23.0	0.8	0.0	1.0	0.0
psychologist	58.6	51.0	22.4	40.8	52.2
receptionist	91.4	99.6	100.0	99.2	99.8
reporter	44.4	53.7	52.5	44.0	57.6
researcher	44.6	77.3	47.8	52.8	55.0
salesperson	39.8	56.8	7.0	37.4	30.5
scientist	33.4	23.0	22.1	15.9	45.3
senator	35.0	22.7	8.0	12.0	12.5
singer	57.6	74.0	54.1	66.6	61.2
student	63.0	44.6	32.3	51.8	40.5
supervisor	65.2	20.9	5.6	18.2	15.0
surgeon	30.2	82.0	20.4	50.8	81.6
teacher	63.0	78.7	58.2	87.4	84.6
technician	31.2	0.4	0.2	1.6	0.0
therapist	74.8	88.5	80.8	82.2	88.7
tutor	59.2	48.8	24.1	24.4	50.4
veterinarian	55.2	65.6	48.9	48.7	89.5
writer	30.2	79.2	34.7	69.1	76.6

Table 7: The percentage of females across occupations in training images (using our initial approach from Section 4) and generated images using **SD 1.5**. We display generation results for each prompt. **Highlighted rows** indicate occupations for which bias switches direction from training to generation across all prompts.